

A Practical Guide to Attention mechanism neural network

TOPIC -- ATTENTION MECHANISM NEURAL NETWORK

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

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Multi-head attention In the self-attention mechanism, queries (Q), keys (K), and values (V) are dynamically generated for each input sequence (typically limited by the size of the context window), allowing the model to focus on different parts of the input sequence at different steps. Multi-head attention enhances this process by introducing multiple parallel attention heads. Each attention head learns different linear projections of the Q, K, and V matrices. This allows the model to capture different aspects of the relationships between words in the sequence simultaneously, rather than focusing on a single aspect. By doing this, multi-head attention ensures that the input embeddings are updated from a more varied and diverse set of perspectives. After the attention outputs from all heads are calculated, they are concatenated and passed through a final linear transformation to generate the output.

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AI boom era As early as spring 2017, even before the "Attention is all you need" preprint was published, one of the co-authors applied the "decoder-only" variation of the architecture to generate fictitious Wikipedia articles. Transformer architecture is now used alongside many generative models that contribute to the ongoing AI boom. The "reference implementation" of the original Transformer was written in a TensorFlow library. In language modelling, ELMo (2018) was a bi-directional LSTM that produces contextualized word embeddings, improving upon the line of research from bag of words and word2vec. It was followed by BERT (2018), an encoder-only transformer model. In October 2019, Google started using BERT to process search queries. In 2020, Google Translate replaced the previous RNN-encoder–RNN-decoder model by a transformer-encoder–RNN-decoder model. Starting in 2018, the OpenAI GPT series of decoder-only transformers became state of the art in natural language generation. At the end of 2022, ChatGPT, a chatbot based on a fine-tuned variant of GPT-3.5, became unexpectedly popular, triggering a boom around

large language models. Transformers have been applied in modalities beyond text. Four days after the publication of "Attention is All You Need", a multimodal transformer architecture, MultiModel, was published by most authors of that paper. Other examples include the vision transformer, speech recognition, robotics, and multimodal. The vision transformer, in turn, stimulated new developments in convolutional neural networks. Image and video generators like DALL-E (2021), Stable Diffusion 3 (2024), and Sora (2024), use transformers to analyse input data (like text prompts) by breaking it down into "tokens" and then calculating the relevance between each token using self-attention, which helps the model understand the context and relationships within the data.

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where Q , K , V are respectively the query, key, value matrices, and d_k is the dimension of the values. Since the model relies on Query (Q), Key (K), and Value (V) matrices that come from the same source (i.e., the input sequence or context window), this eliminates the need for RNNs, completely ensuring parallelizability for the architecture. This differs from the original form of the Attention mechanism introduced in 2014. Additionally, the paper also discusses the use of an additional scaling factor that was found to be most effective with respect to the dimension of the key vectors (represented as d_k and initially set to 64 within the paper) in the manner shown above. In the specific context of translation, which the paper focused on, the Query and Key matrices are usually represented in embeddings corresponding to the source language, while the Value matrix corresponds to the target language.

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wherein p , o , s , i , d , m , o , d , e , l correspond to the position of the word, the current dimension index, and the dimension of the model, respectively. The sine function is used for even indices of the embedding while the cosine function is used for odd indices. The resultant PE embedding is then added to the word at that corresponding position with respect to the current context window. The paper specifically comments on why this method was chosen describing: "We chose the sinusoidal version because it may allow the model to extrapolate to sequence lengths longer than the ones encountered during training."

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The idea of encoder–decoder sequence transduction had been developed in the early 2010s; commonly cited as the originators that produced seq2seq are two concurrently published papers from 2014. A 380M-parameter model for machine translation uses two long short-term memories (LSTM). Its architecture consists of two parts. The encoder is an LSTM that takes in a sequence of tokens and turns it into a vector. The decoder is another LSTM that converts the vector into a sequence of tokens. Similarly, another 130M-parameter model used gated recurrent units (GRU) instead of LSTM. Later research showed that GRUs are neither better nor worse than LSTMs for seq2seq. These early seq2seq models had no attention mechanism, and the state vector is accessible only after the last word of the source text was processed. Although in theory such a vector retains the information about the whole original sentence, in practice the information is poorly preserved. This is because the input is processed sequentially by one recurrent network into a fixed-size output vector, which is then processed by another recurrent network into an output. If the input is long, then the output vector would not be able to contain all relevant information, degrading the output. As evidence, reversing the input sentence improved seq2seq translation. The RNN search model introduced an attention mechanism to seq2seq for machine translation to solve the bottleneck problem (of the fixed-size output vector), allowing the model to process long-distance dependencies more easily. The name is because it "emulates searching through a source sentence during decoding a translation". The relative performances were compared between global (that of RNN search) and local (sliding window) attention model architectures for machine translation, finding that mixed attention had higher quality than global attention, while local attention reduced translation time. In 2016, Google Translate was revamped to Google Neural Machine Translation, which replaced the previous model based on statistical machine translation. The new model was a seq2seq model where the encoder and the decoder were both 8 layers of bidirectional LSTM. It took nine months to develop, and it outperformed the statistical approach, which took ten years to develop.